THE INDIAN INTERNATIONAL SCHOOL,DSO
Worksheet – Chapter 12: Linear Programming (2026-2027)

Grade: 12 Subject: MATHEMATICS Total marks: 27

Name: _____
Date: _____

Instructions:

- This worksheet has 10 questions arranged in three levels of increasing difficulty.
- Level 1: Q 1 – Q 4 (MCQs, Assertion–Reason and short answer questions)
- Level 2: Q 5 – Q 7 (short answer questions)
- Level 3: Q 8 – Q 10 (long answer and case-based questions)
- Marks for each question are given against it. All questions are compulsory.
- The answer key is given at the end of the worksheet.

Q No.	Question	Marks
	LEVEL 1 **Q 1 – Q 4**	**4**
1.	The corner points of the feasible region of an LPP are $(0,2)$, $(3,0)$, $(6,0)$, $(6,8)$ and $(0,5)$. If $Z = 4x + 6y$, then the minimum value of Z occurs at: (A) $(0,2)$ only (B) $(3,0)$ only (C) the mid-point of the line segment joining $(0,2)$ and $(3,0)$ only (D) any point on the line segment joining $(0,2)$ and $(3,0)$	1
2.	The region represented by the system of inequations $3x + y \geq 3$, $2x - y \geq -5$, $x, y \geq 0$ is: (A) unbounded in the 1st quadrant (B) bounded in the 1st quadrant (C) unbounded in the 2nd quadrant (D) bounded in the 2nd quadrant	1
3.	The restrictions imposed on the decision variables involved in the objective function of a linear programming problem are called: (A) feasible solutions (B) constraints (C) optimal solutions (D) infeasible solutions	1

Q No.	Question	Marks
4.	**Assertion (A):** If the feasible region of an LPP is bounded, the objective function $Z = ax + by$ has both a maximum and a minimum value on it. **Reason (R):** If the feasible region is bounded, the maximum and minimum values of Z each occur at a corner point of the feasible region. (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). (B) Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). (C) Assertion (A) is true, but Reason (R) is false. (D) Assertion (A) is false, but Reason (R) is true.	1
	LEVEL 2 **Q 5 – Q 7**	**9**
5.	Solve the following linear programming problem graphically: Minimise $Z = 600x + 400y$ subject to the constraints $x + y \geq 8$, $x + 2y \leq 16$, $4x + y \leq 29$, $x, y \geq 0$.	3
6.	Solve the following LPP graphically: Minimise $Z = 3x + 5y$ subject to $x + 3y \geq 3$, $x + y \geq 2$, $x \geq 0$, $y \geq 0$.	3
7.	Solve the following LPP graphically: Maximise $Z = 5x + 3y$ subject to $3x + 5y \leq 15$, $5x + 2y \leq 10$, $x \geq 0$, $y \geq 0$.	3
	LEVEL 3 **Q 8 – Q 10**	**14**
8.	Solve graphically: Minimise and maximise $Z = x + 2y$ subject to $x + 2y \geq 100$, $2x - y \leq 0$, $2x + y \leq 200$, $x \geq 0$, $y \geq 0$.	5
9.	A furniture dealer deals in only two items, tables and chairs. He has ₹50,000 to invest and storage space for at most 60 pieces. A table costs ₹2,500 and a chair ₹500. He expects a profit of ₹250 on a table and ₹75 on a chair. Formulate this as an LPP and solve it graphically to find how many tables and chairs he should buy to maximise his profit, assuming he can sell all the items he buys.	5
10.	**Case Study:** A dietician wants to prepare a mixture of two foods F_1 and F_2. One kg of F_1 contains 2 units of vitamin A and 1 unit of minerals; one kg of F_2 contains 1 unit of vitamin A and 2 units of minerals. The mixture must contain at least 8 units of vitamin A and 10 units of minerals. F_1 costs ₹50 per kg and F_2 costs ₹70 per kg. Let x kg of F_1 and y kg of F_2 be used. **(i)** Write the objective function and the constraints of the LPP. **[1]** **(ii)** Find the corner points of the feasible region. **[1]** **(iii)** Find the minimum cost of the mixture, and justify why the minimum exists even though the feasible region is unbounded. **[2]**	4

***** End of Worksheet *****

ANSWER KEY

Chapter 12: Linear Programming

Q No.	Answer / Key Steps
	LEVEL 1
1.	(D) $Z(0,2) = Z(3,0) = 12$ is the least value, so the minimum occurs at every point of the segment joining them.
2.	(A) unbounded in the 1st quadrant
3.	(B) constraints
4.	(A) Both true; R explains A.
	LEVEL 2
5.	Corner points $(0,8)$, $(6,5)$, $(7,1)$; $Z = 3200,5600,4600$. Minimum $Z = 3200$ at $(0,8)$.
6.	Corner points $(3,0)$, $\left(\frac{3}{2},\frac{1}{2}\right)$, $(0,2)$; $Z = 9,7,10$. Region is unbounded; $3x + 5y < 7$ has no point in common with it, so minimum $Z = 7$ at $\left(\frac{3}{2},\frac{1}{2}\right)$.
7.	Corner points $(0,0)$, $(2,0)$, $\left(\frac{20}{19},\frac{45}{19}\right)$, $(0,3)$; $Z = 0,10,\frac{235}{19},9$. Maximum $Z = \frac{235}{19}$ at $\left(\frac{20}{19},\frac{45}{19}\right)$.
	LEVEL 3
8.	Corner points $(0,50)$, $(20,40)$, $(50,100)$, $(0,200)$; $Z = 100,100,250,400$. Minimum 100 at every point of the segment joining $(0,50)$ and $(20,40)$; maximum 400 at $(0,200)$.
9.	Maximise $Z = 250x + 75y$ subject to $x + y \leq 60$, $5x + y \leq 100$, $x, y \geq 0$. Corner points $(0,0)$, $(20,0)$, $(10,50)$, $(0,60)$; $Z = 0,5000,6250,4500$. Buy 10 tables and 50 chairs; maximum profit ₹6,250.
10.	(i) Minimise $Z = 50x + 70y$ subject to $2x + y \geq 8$, $x + 2y \geq 10$, $x, y \geq 0$. (ii) $(0,8)$, $(2,4)$, $(10,0)$. (iii) $Z = 560,380,500$. The half-plane $50x + 70y < 380$ has no point in common with the feasible region, so the minimum cost is ₹380 with 2 kg of F_1 and 4 kg of F_2.